

Enterprise Architecture Guide Project at MIT

CSG Presentation 9/23/04
Jerry Grochow

MIT's ITAG team sponsored a six week initiative to document the current state architecture and to begin development of a vision for future state enterprise architecture.

Documents will be on-line, refined, and continuously updated to reflect deeper understanding and changing needs of the Institute.

Scope of Enterprise Architecture:

- Business processes and products, applications and data (Process)
- Software and hardware infrastructure (Technology)
- Knowledge and expertise (People)

Current State Assessment:

- Documentation review
- Interviews with ITAG and departmental technology staff
- Workshops to validate information gathered

Future State:

- Workshops (1/2 day) to cover:
 - Future State Context
 - Future State Principles
 - Logical Architecture Framework
 - Technology Standards
 - Services Matrix
 - Project Review process

Future Phases:

- Roadmap and Planning
- Build-out

Project Sponsors and Departmental Leadership

- Understand institute investments in technology
- Understand architectural governance process
- Provide technical direction to guide their investments
- Provide a way to assess architectural risks on projects

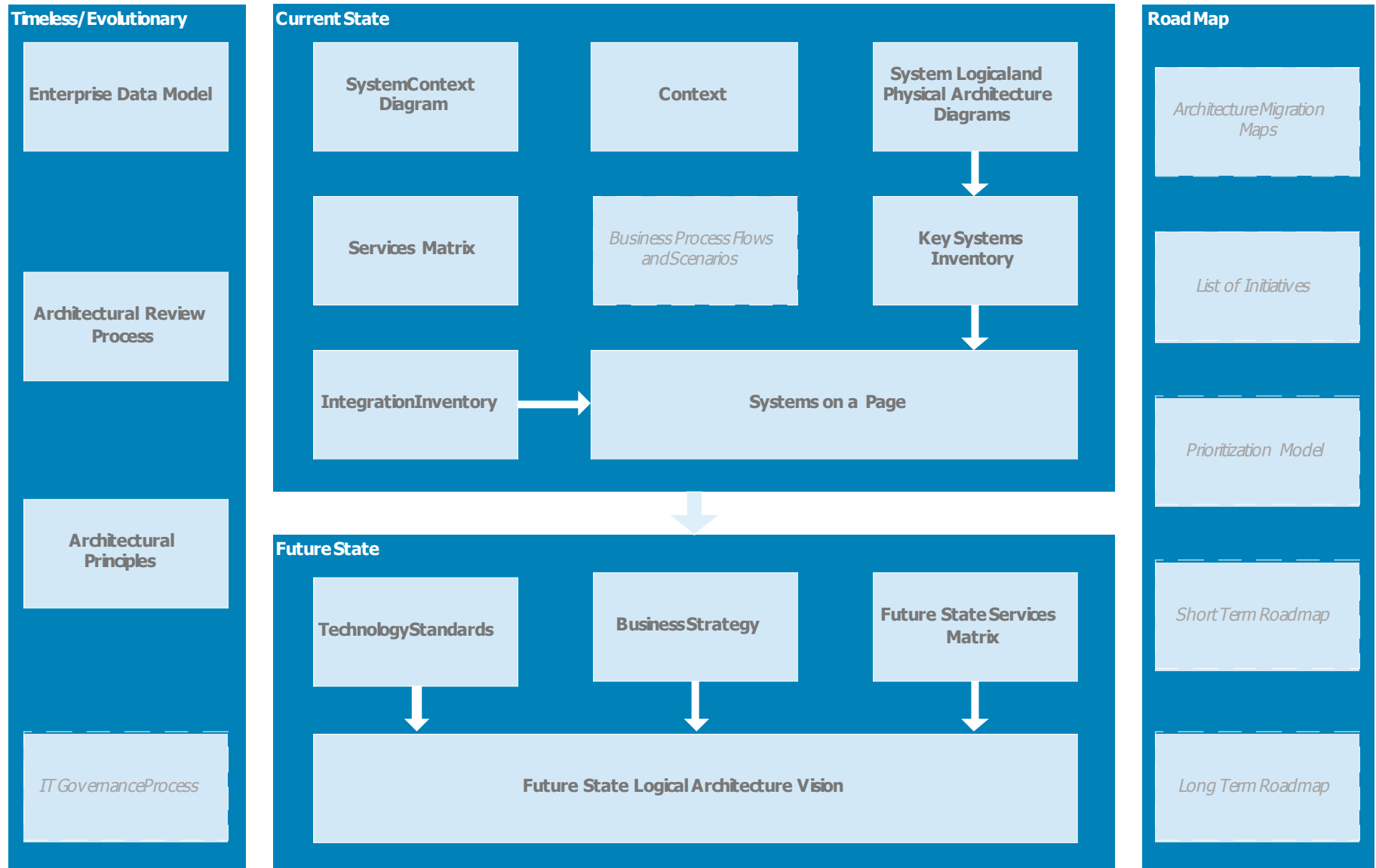
IT Architecture Group

- Common understanding of architecture
- Frames architectural review process for projects
- Inform gaps in the architecture and where we need to evolve
- Provide a way to assess architectural risks on projects

Developers

- Awareness
 - of available services
 - of enterprise data model
 - of the landscape and future vision of enterprise architecture
 - of supported and recommended standards
- Reduce redundancy of effort
- Provide a way to assess architectural risks on projects
- Know where to go to get help
- Awareness of their role in expanding the services offered in the enterprise architecture

Enterprise Architecture Deliverables



Future State Context



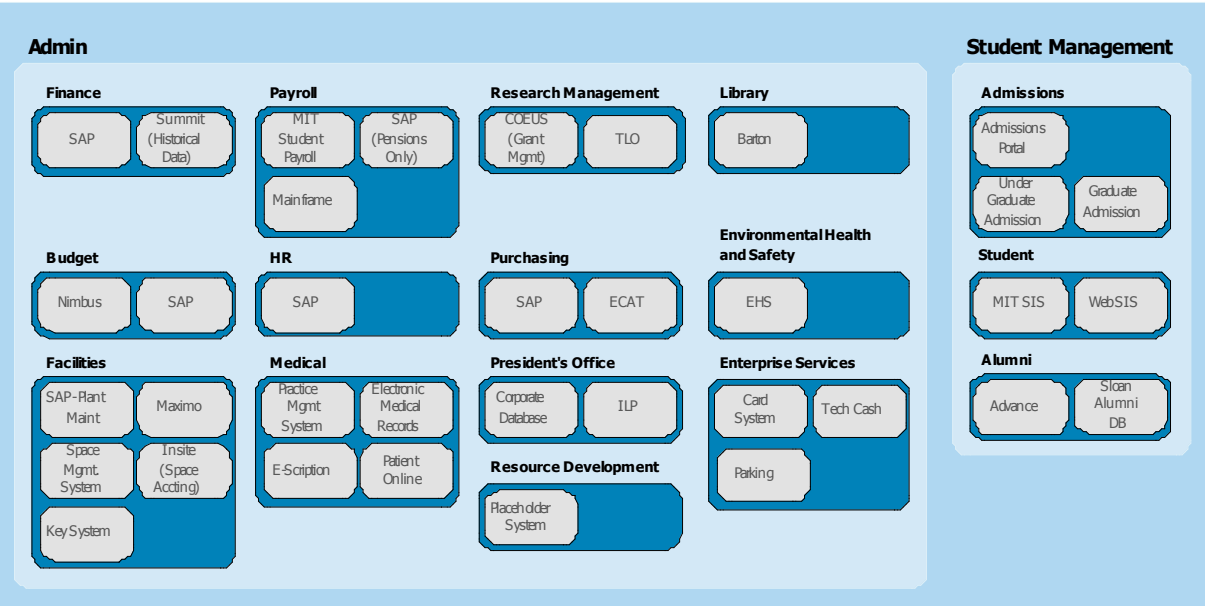
- SAP will continue to be the primary ERP system
- The MIT Data Warehouse will be the central repository for administrative data that is of interest to more than one DLC.
- The MIT community will continue to be global, 24x7 and evolving
- There will be increased integration between MIT and other universities
- The MIT environment is heterogeneous
- The MIT network will evolve to support needs of the enterprise
 - We may have many research networks
 - We will have an IPv6 network and we will need differentiated services to better support user needs

- **Security:** applications should ensure data and access security
- **Ownership:** clear and explicit ownership of enterprise data
- **Leverage assets:** leverage existing services and capabilities
- **Accessibility:** be aware of to needs of all users (location & disabilities)
- **Read-time:** Minimize latency of data updates
- **Standards:** promote consistency using standards

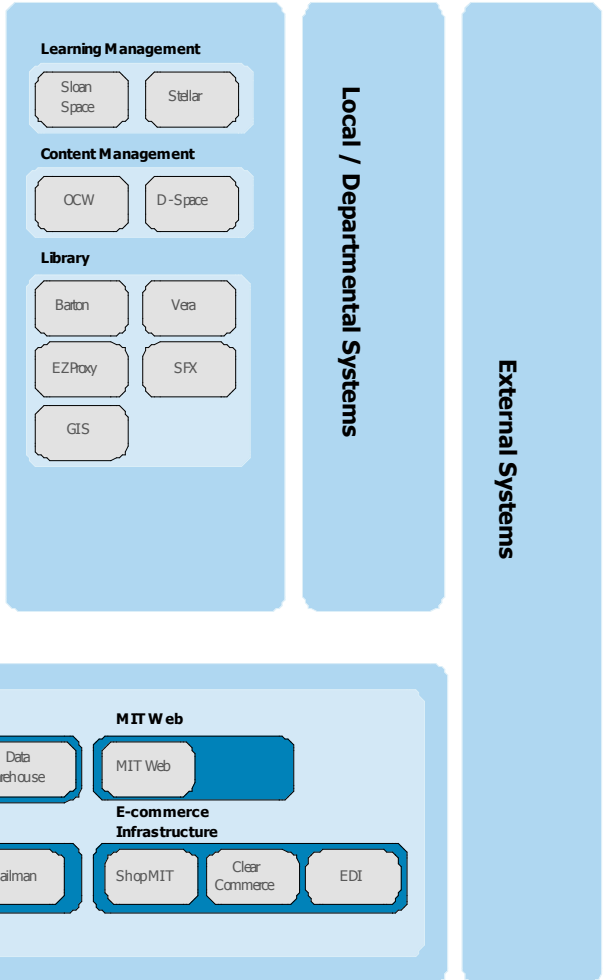
Systems Context Diagram (Vers. 0.1)



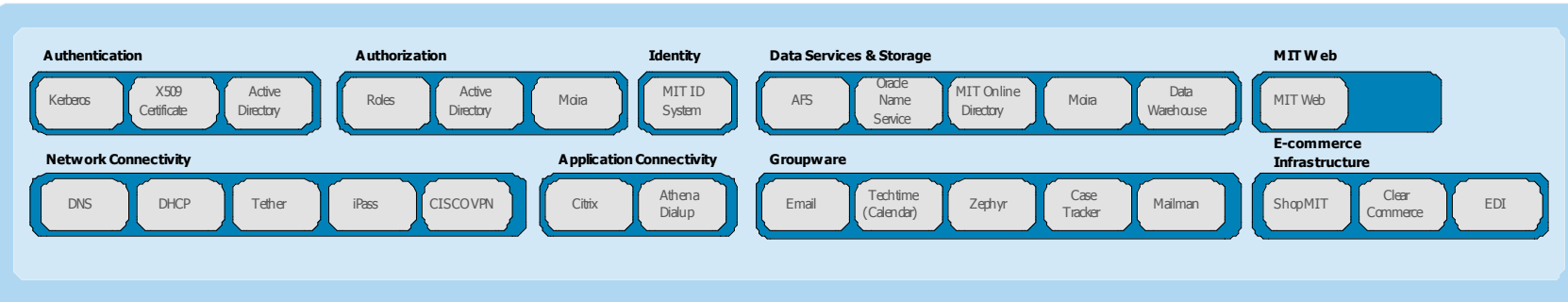
Enterprise Administrative Systems



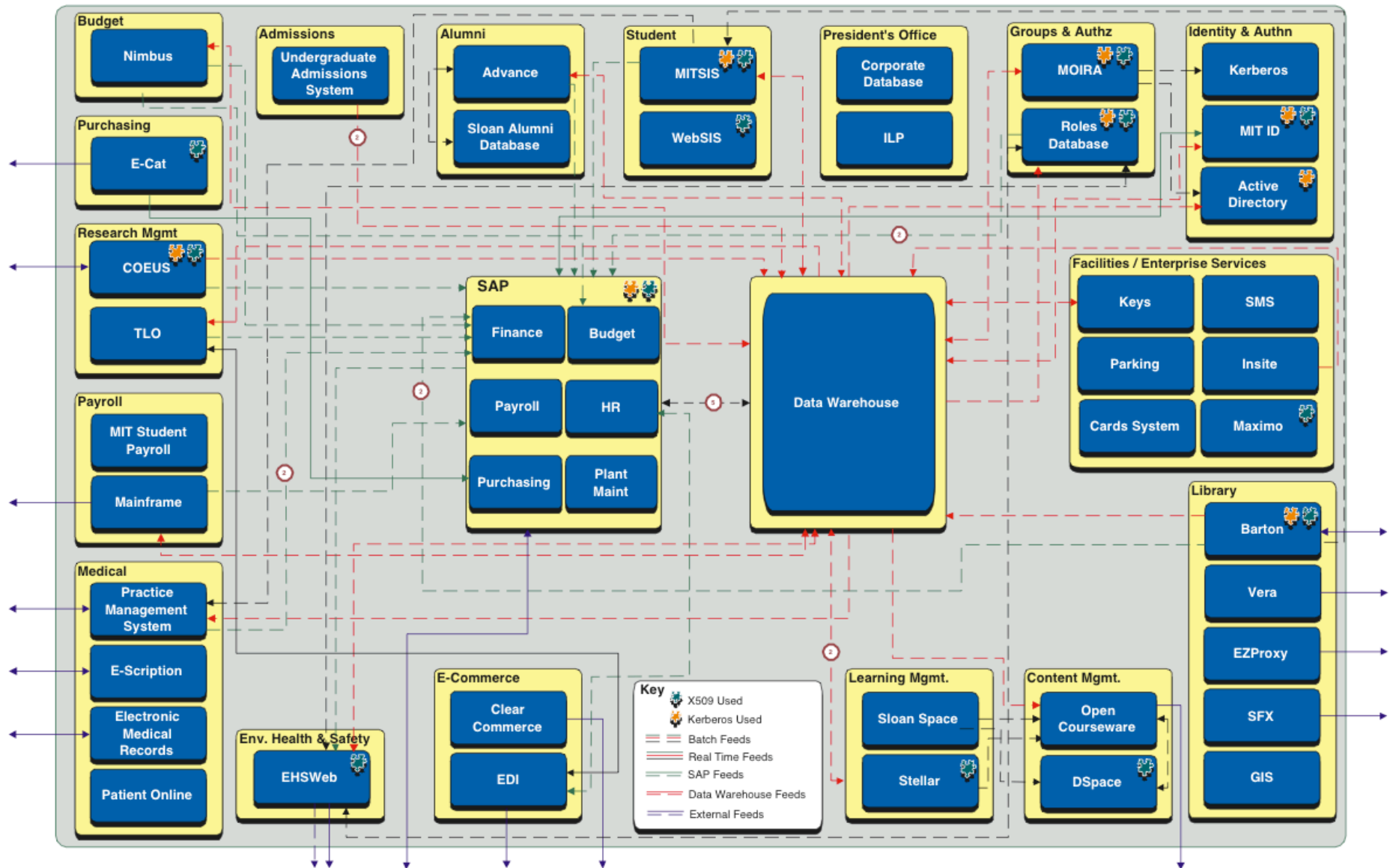
Academic Systems

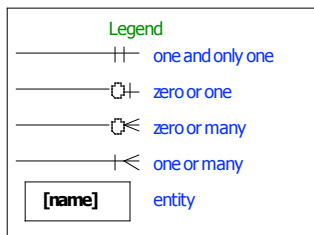


Integrated Operation/Infrastructure Support

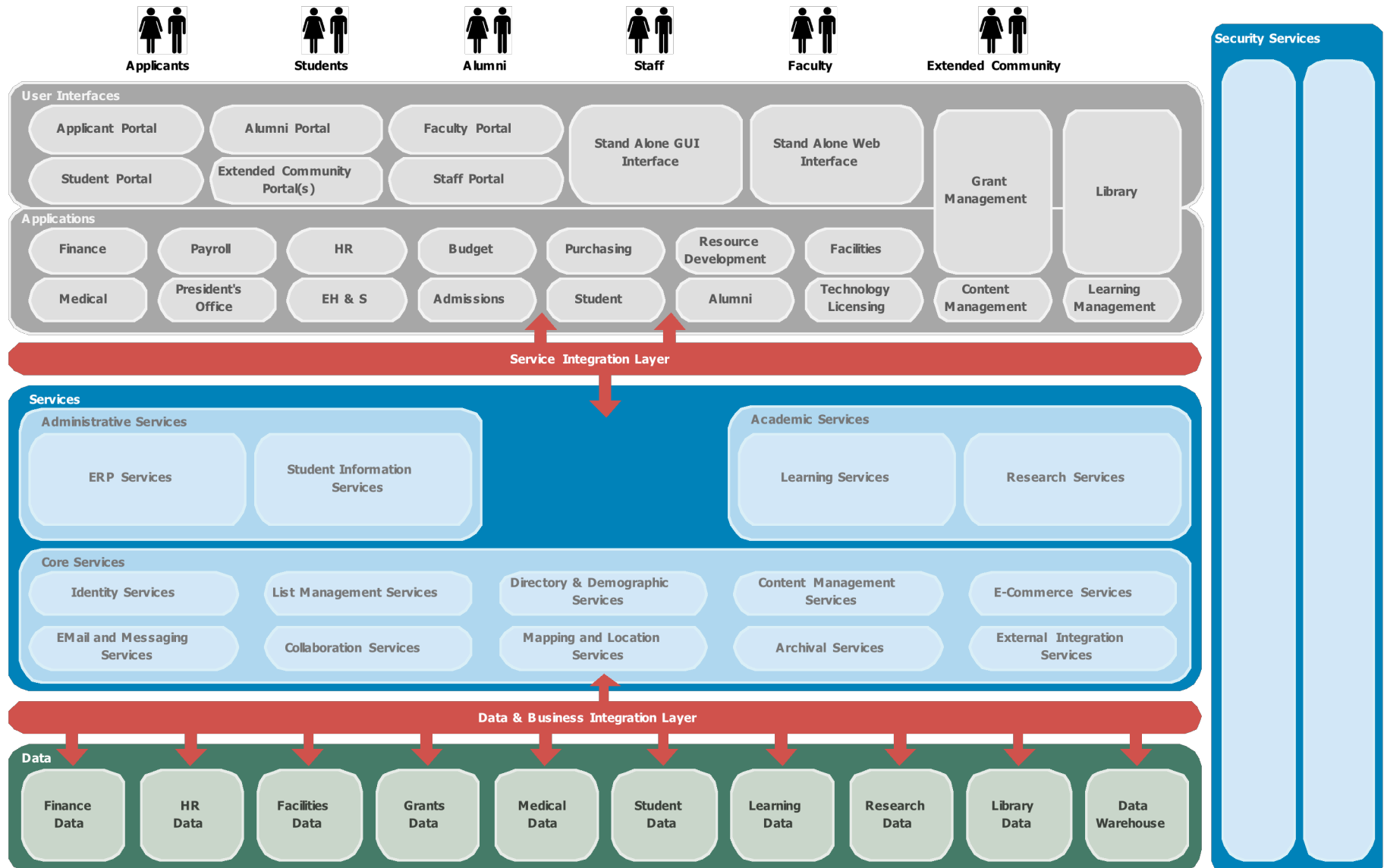


Systems on a Page (Vers. 0.1)

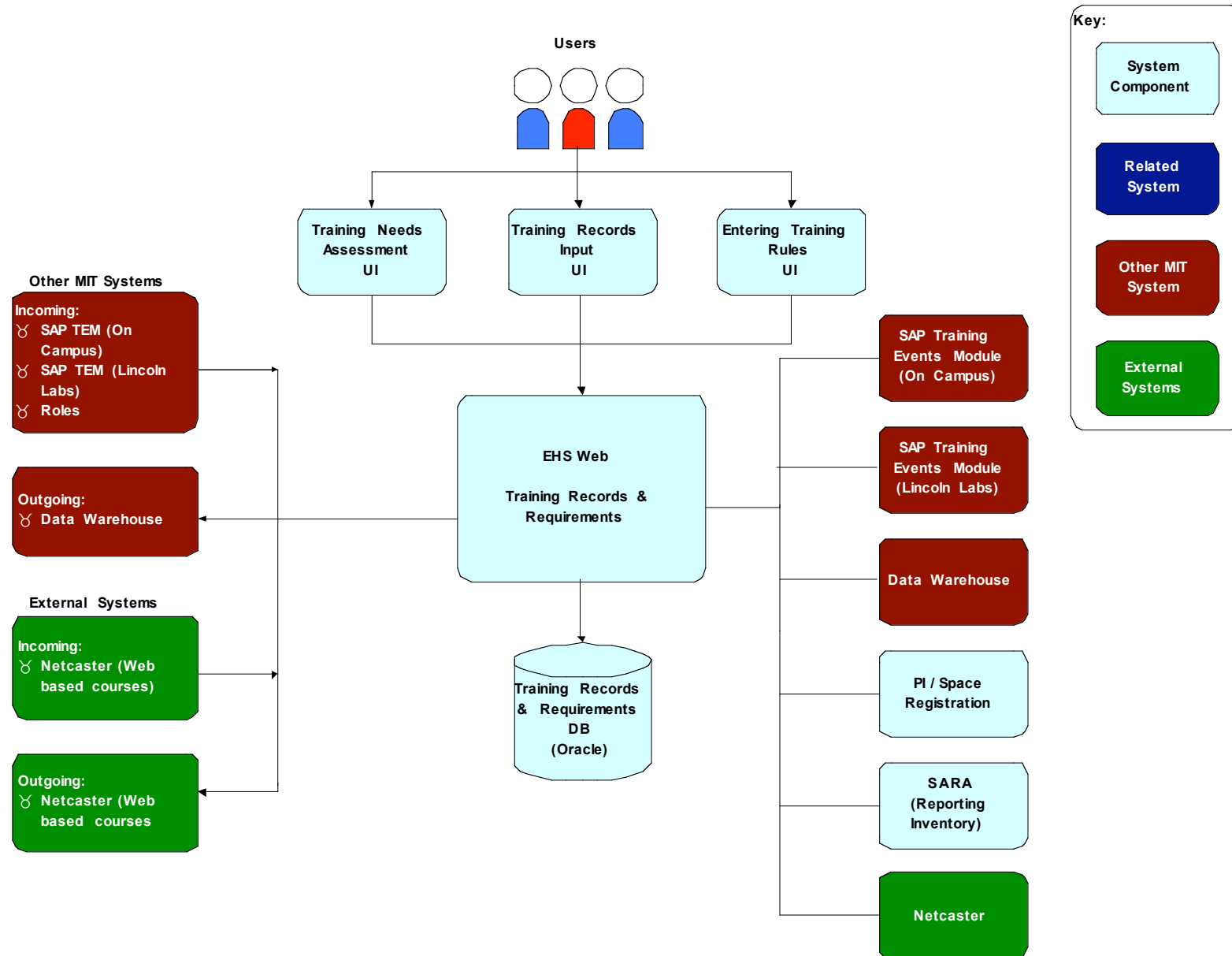




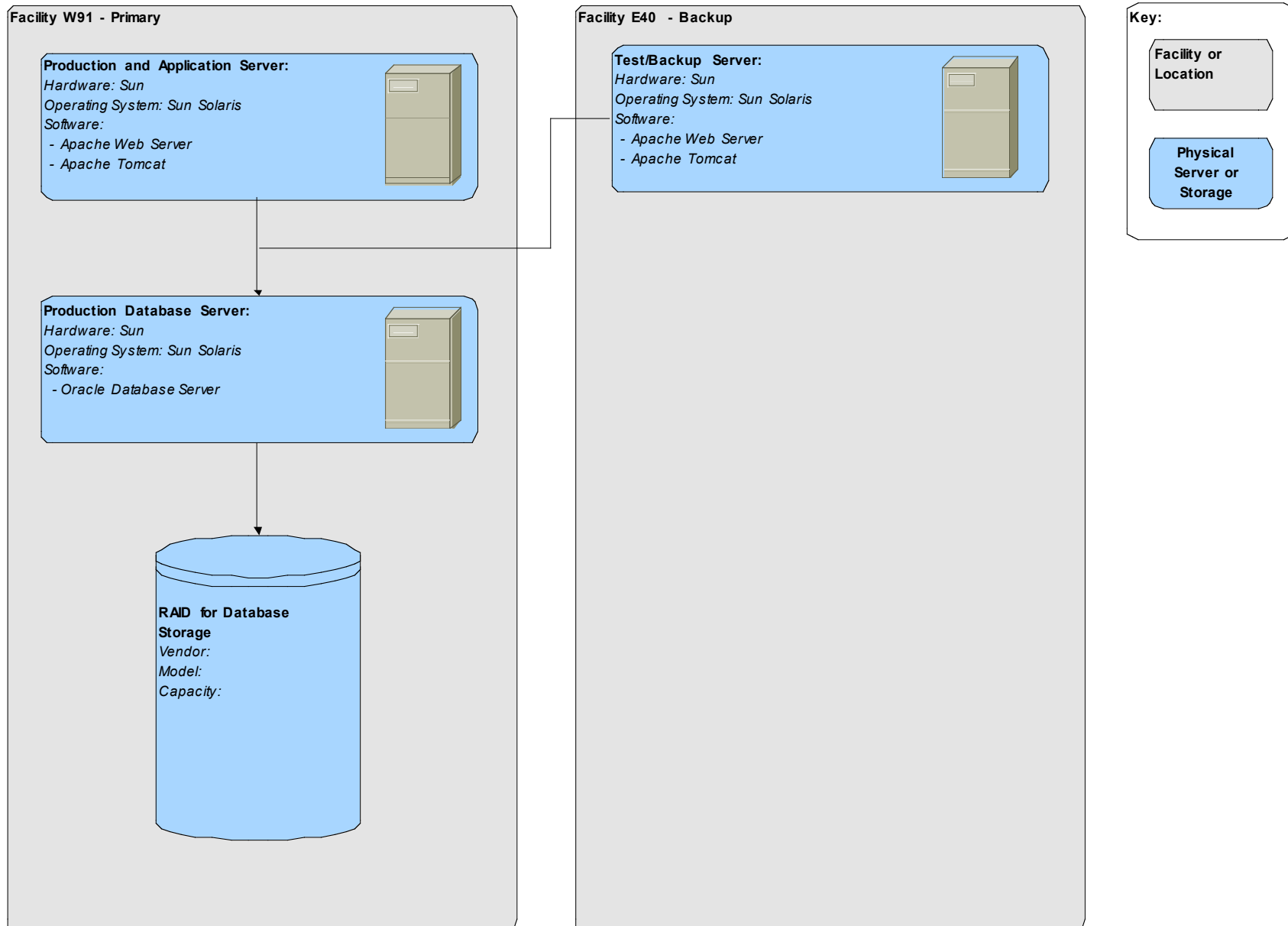
Future State Vision (Vers. 0.1)



Example Logical Architecture Diagram: EHSWeb



Example Physical Architecture Diagram: EHSWeb



MIT Enterprise Architecture Guide Project

Further information:

Jerry Grochow (jgrochow@mit.edu)

Scott Thorne (thorne@mit.edu)

Steve Winig (swinig@mit.edu)